

Database Testing Report

E-Commerce Graduation Project | DEPI Program

Team	Team B
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Database	TrelloTestingDB (MySQL)
Tool	MySQL Workbench
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Overall Summary

38
Total Test Cases

32
Pass

6
Expected to Fail (Constraints)

100%
Coverage

🚩 The 6 constraint test cases (DB_TC_12 to DB_TC_17) are designed to produce database errors on purpose. Receiving the expected error code is the passing condition for those cases. Each one is run separately in MySQL Workbench.

Section 1 – Schema Verification | DB_TC_01 to DB_TC_05

Test ID	Test Name	What We Check	Expected Result	Status
DB_TC_01	All tables exist	SHOW TABLES	Boards, Lists, Cards, TestCases all appear	PASS
DB_TC_02	Boards table structure	DESCRIBE Boards	BoardID (PK), BoardName (NOT NULL), Description	PASS
DB_TC_03	Lists table structure	DESCRIBE Lists	ListID (PK), ListName (NOT NULL), BoardID (FK)	PASS
DB_TC_04	Cards table structure	DESCRIBE Cards	CardID (PK), CardName (NOT NULL), ListID (FK), Status	PASS
DB_TC_05	TestCases table structure	DESCRIBE TestCases	TestCaseID (AUTO_INCREMENT PK), Method, Endpoint, StatusCode, Status	PASS

Section 2 – Insert Tests (CREATE) | DB_TC_06 to DB_TC_11

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_06	Insert valid Board	INSERT into Boards with full valid data	1 row affected, no error	PASS
DB_TC_07	Insert valid List	INSERT into Lists linked to a valid BoardID	1 row affected, no error	PASS
DB_TC_08	Insert valid Card	INSERT into Cards linked to a valid ListID	1 row affected, no error	PASS
DB_TC_09	Insert Card with empty CardName	INSERT Card where CardName = "	1 row affected (empty string is not NULL)	PASS
DB_TC_10	Insert Card with special characters	INSERT Card with @#! & " in CardName	1 row affected, characters stored exactly as entered	PASS
DB_TC_11	Batch insert TestCases	INSERT 5 rows covering POST / GET / PUT / DELETE / 404	5 rows affected, no error	PASS

Section 3 – Constraint Violation Tests | DB_TC_12 to DB_TC_17

These tests are run one by one. The error message received IS the passing condition.

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_12	Duplicate Primary Key	INSERT a Card using an existing CardID	ERROR 1062 – Duplicate entry for PRIMARY KEY	ERROR 1062 ✓
DB_TC_13	NULL CardName rejected	INSERT a Card with CardName = NULL	ERROR 1048 – Column CardName cannot be null	ERROR 1048 ✓
DB_TC_14	Invalid ListID rejected	INSERT a Card with a ListID that does not exist	ERROR 1452 – Foreign key constraint fails	ERROR 1452 ✓
DB_TC_15	Invalid BoardID rejected	INSERT a List with a BoardID that does not exist	ERROR 1452 – Foreign key constraint fails	ERROR 1452 ✓
DB_TC_16	Delete List blocked by Cards	DELETE a List while Cards still reference it	ERROR 1451 – Cannot delete, Cards reference this List	ERROR 1451 ✓
DB_TC_17	Delete Board blocked by Lists	DELETE a Board while Lists still reference it	ERROR 1451 – Cannot delete, Lists reference this Board	ERROR 1451 ✓

Section 4 – Select Tests (READ) | DB_TC_18 to DB_TC_24

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_18	Retrieve all Cards	SELECT * FROM Cards	All inserted cards appear in the result	PASS
DB_TC_19	Retrieve Card by ID	SELECT WHERE CardID = valid ID	Exactly 1 row returned for Test Card	PASS
DB_TC_20	Retrieve non-existent Card	SELECT WHERE CardID = fake ID	0 rows returned	PASS
DB_TC_21	Retrieve passing TestCases	SELECT WHERE Status = 'Pass'	All 5 TestCase rows returned	PASS
DB_TC_22	JOIN Cards and Lists	JOIN Cards on Lists using ListID	Each card row shows its correct ListName	PASS
DB_TC_23	Full hierarchy JOIN	3-table JOIN across Board, List, Card	BoardName, ListName, CardName shown in one row per card	PASS
DB_TC_24	Card count per List	SELECT COUNT(*) GROUP BY ListID	ListID 689006 shows 3 cards	PASS

Section 5 – Update Tests | DB_TC_25 to DB_TC_30

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_25	Update CardName	SET CardName = 'Grad Project'	CardName updated to 'Grad Project'	PASS
DB_TC_26	Update Description	SET Description = new text	Description updated correctly	PASS
DB_TC_27	Update CardName to empty string	SET CardName = "	CardName becomes empty string, no error	PASS
DB_TC_28	Update non-existent Card	UPDATE WHERE CardID = fake ID	0 rows affected, no error	PASS
DB_TC_29	Partial update – Status only	SET Status = 'Archived'	Only Status changes, all other fields unchanged	PASS
DB_TC_30	Soft-delete flag	SET Status = 'Deleted'	Status becomes 'Deleted'	PASS

Section 6 – Delete Tests | DB_TC_31 to DB_TC_33

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_31	Delete existing Card	DELETE WHERE CardID = valid ID	1 row affected, no error	PASS
DB_TC_32	Confirm Card is gone	SELECT after DELETE	0 rows returned	PASS
DB_TC_33	Delete non-existent Card	DELETE WHERE CardID = fake ID	0 rows affected, no error	PASS

Section 7 – Data Integrity Checks | DB_TC_34 to DB_TC_38

Test ID	Test Name	What We Do	Expected Result	Status
DB_TC_34	No orphaned Cards	LEFT JOIN Cards → Lists, check for NULL ListID	0 rows – no broken references	PASS
DB_TC_35	No orphaned Lists	LEFT JOIN Lists → Boards, check for NULL BoardID	0 rows – no broken references	PASS
DB_TC_36	POST record correct	SELECT POST row from TestCases	Method = POST, StatusCode = 200, Status = Pass	PASS
DB_TC_37	DELETE record correct	SELECT DELETE row from TestCases	Method = DELETE, StatusCode = 200, Status = Pass	PASS
DB_TC_38	404 record correct	SELECT row WHERE StatusCode = 404	StatusCode = 404, Status = Pass	PASS